ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Section A: objective questions (30) · Section B: technical drawings and symbols, measuring instruments, materials, fasteners and joining, mechanisms and mechanical calculations, open- and closed-loop control, and basic electrical control (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. Use g = 9,81 m/s² where needed.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 In first-angle orthographic projection, the top view is drawn …

A above the front view. B below the front view.
C to the left of the front view. D on a separate page only.

1.1.2 The instrument that measures the diameter of a shaft to 0,01 mm is a …

A steel rule. B tape measure.
C try square. D micrometer.

1.1.3 A material that returns to its original shape after a force is removed is …

A brittle. B elastic. C plastic. D ductile.

1.1.4 A fastener that can be removed and used again is a …

A rivet. B weld. C nut and bolt. D glued joint.

1.1.5 A cam and follower changes …

A rotary motion into reciprocating motion. B electrical energy into heat.
C AC into DC. D linear motion into sound.

1.1.6 A driver gear has 20 teeth and the driven gear has 60 teeth. The gear ratio (driven : driver) is …

A 1 : 3 B 3 : 1 C 40 : 1 D 1 : 40

1.1.7 A system that does not use feedback is a/an …

A closed-loop system. B servo system.
C PID system. D open-loop system.

1.1.8 In a control system, the device that measures the output and sends the information back is the …

A sensor (feedback element). B actuator.
C controller. D power supply.

1.1.9 A switch that stays in its new position after being pressed is a …

A push-to-make (momentary) switch. B limit switch only.
C latching switch. D reed relay.

1.1.10 The unit of work and energy is the …

A watt. B newton. C joule. D pascal.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Hidden edges on a technical drawing are shown with short dashed lines.

1.2.2 A vernier caliper can measure inside and outside diameters and depths.

1.2.3 Aluminium is lighter than steel.

1.2.4 Spring washers help stop nuts from loosening because of vibration.

1.2.5 A lever with a mechanical advantage greater than 1 makes the load harder to lift.

1.2.6 A rack and pinion changes rotary motion into linear motion.

1.2.7 A thermostat in a geyser is an example of closed-loop control.

1.2.8 A toaster with a timer only is an example of closed-loop control.

1.2.9 A relay allows a small current to switch a larger current.

1.2.10 Power is the rate at which work is done.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Line used to show the centre of a hole or shaft	A	work
1.3.2	Allowed difference between the largest and smallest size of a part	B	limit switch
1.3.3	Ability of a material to be drawn into wire	C	fulcrum
1.3.4	Ability to resist scratching and wear	D	mechanical advantage
1.3.5	Permanent joint made by melting metal	E	feedback
1.3.6	Fixed point about which a lever turns	F	hardness
1.3.7	Load ÷ effort	G	ductility
1.3.8	Signal sent back from the output to the controller	H	weld
1.3.9	Switch operated by a moving part reaching the end of its travel	I	centre line

1.3.10	Force × distance moved in the direction of the force	J	tolerance

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: DRAWINGS AND MEASURING

2.1 Name the THREE main views in an orthographic drawing and state what each view shows. (6)

2.2 Draw and name FOUR types of lines used on technical drawings. (4)

2.3 A vernier caliper has a main-scale reading of 23 mm and the 14th vernier line lines up (0,02 mm per division). Calculate the reading. (3)

2.4 Give THREE rules for the care of precision measuring instruments. (3)

[16]

QUESTION 3: MATERIALS, FASTENERS AND JOINING

3.1 Recommend a material for EACH robot part and give a reason: (a) the frame of a light robot arm; (b) a gear that must be quiet and needs no lubrication; (c) the cover of an electronic control box. (6)

3.2 Differentiate between permanent and temporary joining methods, and give TWO examples of each. (6)

3.3 Explain why a locknut or thread-locking fluid is used on a moving machine. (2)

[14]

QUESTION 4: MECHANISMS AND CALCULATIONS

4.1 A first-class lever is 1,2 m long with the fulcrum 0,3 m from a 60 kg load.

4.1.1 Calculate the weight of the load. (2)

4.1.2 Calculate the effort needed at the other end to balance the load. (3)

4.1.3 Calculate the mechanical advantage. (2)

4.2 A motor turns a 20-tooth gear at 1 500 r/min. It drives a 75-tooth gear.

4.2.1 Calculate the speed of the driven gear. (3)

4.2.2 State whether the output torque is larger or smaller than the input torque, and explain why. (2)

4.3 A robot lifts a 5 kg box 0,8 m in 2 seconds. Calculate the work done and the power. (5)

4.4 Name TWO mechanisms that change rotary motion into linear motion and give a robot example of each. (4)

[21]

QUESTION 5: CONTROL AND ELECTRICAL CONTROL

5.1 Differentiate between open-loop and closed-loop control, and give ONE robot example of each. (6)

5.2 Draw a block diagram of a closed-loop system that keeps a robot's motor at a set speed, and label the input, controller, actuator, process, sensor and feedback. (6)

5.3 Explain the function of a limit switch on a robot's linear axis. (2)

5.4 Give FIVE safety rules for working on electrical control panels. (5)

[19]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER